

Lecture 5: Kalman Filter in nD and Some Nonlinear Filters

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Introduction to Data Assimilation
2026 Spring



A C.C. Lin Distinguished Talk

周五第二讲 | ICM 2014 报告人

Seung-Yeal Ha 院士主讲冠名讲座

YMSC 清华大学丘成桐数学科学中心

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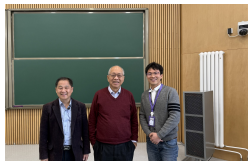
Chia-Chiao Lin Memorial Lectures

— 林家翘冠名讲座 —

Speaker: Seung Yeal Ha



Professor Seung Yeal Ha is a



A C.C. Lin Distinguished Talk

Time: March 27, 2:00 PM - 3:00 PM

Location: Old Economics and Management Lecture Hall, Zoom
Meeting ID: 4552601552 Passcode: YMSC

Emergent dynamics of the inertial spin model on Euclidean spaces

In this talk, we discuss qualitative and quantitative flocking estimates for the inertial spin model on the three-dimensional Euclidean space and its high-dimensional generalization. First, we present several mathematical frameworks leading to the exponential decay of the inertial spin model. For this, we derive a system of differential inequalities for relative kinetic and spin energies, and then we explicitly obtain exponential flocking under the proposed frameworks.

Second, we propose a high-dimensional inertial spin(HIS) model which generalizes the three-dimensional inertial spin model and study its emergent behaviors. For the generalization of the IS model to the high-dimensional Euclidean space, we replace the cross product by the multiplication via skew-symmetric matrix, and identify a new set of constants of motions which are conserved along the proposed model. We provide two sufficient frameworks leading to the collective behaviors of the HIS model leading to qualitative and quantitative emergent dynamics in terms of system parameters and initial data.

Outline

1. Kalman filter in nD
2. Main ideas of several nonlinear filters

1D Kalman Filter Recall

$$\text{Forecast: } u_{m+1} = F u_m + \sigma_{m+1}, \quad \sigma_{m+1} \sim \mathcal{N}(0, r).$$

$$\text{Observation: } v_{m+1} = g u_{m+1} + \sigma_{m+1}^o, \quad \sigma_{m+1}^o \sim \mathcal{N}(0, r^o).$$

The Kalman Filter Algorithm: Given $(\bar{u}_{m|m}, r_{m|m})$:

$$\text{Forecast: } \bar{u}_{m+1|m} = F \bar{u}_{m|m}, \quad r_{m+1|m} = F r_{m|m} F^* + r.$$

$$\text{Analysis: } K_{m+1} = \frac{g r_{m+1|m}}{r^o + g^2 r_{m+1|m}},$$

$$\bar{u}_{m+1|m+1} = \bar{u}_{m+1|m} + K_{m+1} (v_{m+1} - g \bar{u}_{m+1|m}),$$

$$r_{m+1|m+1} = (1 - K_{m+1} g) r_{m+1|m}.$$

The N -dimensional Kalman filter

Our goal is to obtain the posterior distribution $p(\vec{u}|\vec{v}_{m+1})$ through the Bayesian formula assuming a knowledge of the prior distribution $p(\vec{u})$ of a true signal $\vec{u}_{m+1} \in \mathbb{R}^N$ and the observed value $\vec{v}_{m+1} \in \mathbb{R}^M$.

The model is

$$\vec{u}_{m+1|m} = F\vec{u}_{m|m} + \vec{\sigma}_{m+1}, \quad R = \langle \vec{\sigma}_m \otimes \vec{\sigma}_m^T \rangle.$$

The observational process is

$$\vec{v}_{m+1} = G\vec{u}_{m+1} + \vec{\sigma}_{m+1}^o, \quad R^o = \langle \vec{\sigma}_m^o \otimes (\vec{\sigma}_m^o)^T \rangle.$$

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The posterior mean and posterior covariance are given by

$$\begin{aligned} \vec{u}_{m+1|m+1} &= \vec{u}_{m+1|m} + K_{m+1}(\vec{v}_{m+1} - G\vec{u}_{m+1|m}), \\ R_{m+1|m+1} &= (I - K_{m+1}G)R_{m+1|m} \end{aligned}$$

where the Kalman gain is

$$K_{m+1} = R_{m+1|m}G^T(GR_{m+1|m}G^T + R^o)^{-1}.$$

Time Evolution of the Posterior Statistics

In general, the Kalman filter system is given by

$$\mathbf{u}_I(t_{i+1}) = \mathbf{G}\mathbf{u}_{II}(t_{i+1}) + \mathbf{B}_1\varepsilon_1,$$

$$\mathbf{u}_{II}(t_{i+1}) = \mathbf{a}_0 + \mathbf{a}_1\mathbf{u}_{II}(t_i) + \mathbf{b}_2\varepsilon_2,$$

Define $\mathbf{A}_0 = \mathbf{G}\mathbf{a}_0$, $\mathbf{A}_1 = \mathbf{G}\mathbf{a}_1$ and $\mathbf{B}_2 = \mathbf{G}\mathbf{b}_2$. Then the above system becomes

$$\mathbf{u}_I(t_{i+1}) = \mathbf{A}_0 + \mathbf{A}_1\mathbf{u}_{II}(t_i) + \mathbf{B}_1\varepsilon_1 + \mathbf{B}_2\varepsilon_2,$$

$$\mathbf{u}_{II}(t_{i+1}) = \mathbf{a}_0 + \mathbf{a}_1\mathbf{u}_{II}(t_i) + \mathbf{b}_2\varepsilon_2,$$

The time evolutions of the posterior mean and posterior covariance are given by

$$\begin{aligned}\boldsymbol{\mu}(t_{i+1}) &= \mathbf{a}_0 + \mathbf{a}_1\boldsymbol{\mu}(t_i) + (\mathbf{b}_2\mathbf{B}_2^* + \mathbf{a}_1\mathbf{R}(t_i)\mathbf{A}_1^*) \times \\ &\quad (\mathbf{B}_1\mathbf{B}_1^* + \mathbf{B}_2\mathbf{B}_2^* + \mathbf{A}_1\mathbf{R}(t_i)\mathbf{A}_1^*)^{-1} (\mathbf{u}_I(t_{i+1}) - \mathbf{A}_0 - \mathbf{A}_1\boldsymbol{\mu}(t_i)), \\ \mathbf{R}(t_{i+1}) &= \mathbf{a}_1\mathbf{R}(t_i)\mathbf{a}_1^* + \mathbf{b}_2\mathbf{b}_2^* - (\mathbf{b}_2\mathbf{B}_2^* + \mathbf{a}_1\mathbf{R}(t_i)\mathbf{A}_1^*) \times \\ &\quad (\mathbf{B}_1\mathbf{B}_1^* + \mathbf{B}_2\mathbf{B}_2^* + \mathbf{A}_1\mathbf{R}(t_i)\mathbf{A}_1^*)^{-1} (\mathbf{b}_2\mathbf{B}_2^* + \mathbf{a}_1\mathbf{R}(t_i)\mathbf{A}_1^*)^*.\end{aligned}$$

The equation of $\mathbf{R}$ is called the Riccati equation.

What's new in N -dimensional Kalman filter?

Case 1. Model:

$$u_{m+1}^1 = F_{11}u_m^1 + F_{12}u_m^2 + \sigma_m^1$$

$$u_{m+1}^2 = F_{21}u_m^1 + F_{22}u_m^2 + \sigma_m^2$$

Observation:

$$v_{m+1}^1 = g_1 u_{m+1}^1 + \sigma_m^{o,1}$$

$$v_{m+1}^2 = g_2 u_{m+1}^2 + \sigma_m^{o,2}$$

Case 2. Model:

$$u_{m+1}^1 = F_{11}u_m^1 + F_{12}u_m^2 + \sigma_m^1$$

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Case 3. Model:

$$u_{m+1}^1 = F_{11}u_m^1 + F_{12}u_m^2 + \sigma_m^1$$

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Observation:

$$v_{m+1}^1 = g_1 u_{m+1}^1 + g_2 u_{m+1}^2 + \sigma_m^{o,1}$$

Can we filter u_{m+1}^1 ? What about filtering u_{m+1}^2 ?

What's new in N -dimensional Kalman filter?

Case 4. Model:

$$\begin{aligned}u_{m+1}^1 &= F_{11}u_m^1 + \sigma_m^1 \\u_{m+1}^2 &= F_{21}u_m^1 + F_{22}u_m^2 + \sigma_m^2\end{aligned}$$

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Case 5. Model:

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Observation:

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Case 6. Model:

$$\begin{aligned}u_{m+1}^1 &= F_{11}u_m^1 + \sigma_m^1 \\u_{m+1}^2 &= F_{22}u_m^2 + \sigma_m^2\end{aligned}$$

Observation:

$$v_{m+1}^1 = g_1 u_{m+1}^1 + \sigma_m^{o,1}$$

Can we filter u_{m+1}^1 ? What about filtering u_{m+1}^2 ?

Kalman Filter in nD

$$\text{Forecast: } \mathbf{u}_{m+1} = \mathbf{A} \mathbf{u}_m + \mathbf{f}_m + \boldsymbol{\sigma}_m, \quad \boldsymbol{\sigma}_m \sim \mathcal{N}(\mathbf{0}, \mathbf{R}).$$

$$\text{Observation: } \mathbf{v}_{m+1} = \mathbf{G} \mathbf{u}_{m+1} + \boldsymbol{\sigma}_{m+1}^o, \quad \boldsymbol{\sigma}_{m+1}^o \sim \mathcal{N}(\mathbf{0}, \mathbf{R}^o).$$

The KF Algorithm: Given the analysis $(\bar{\mathbf{u}}_{m|m}, \mathbf{R}_{m|m})$ at time m

$$\text{Forecast: } \bar{\mathbf{u}}_{m+1|m} = \mathbf{A} \bar{\mathbf{u}}_{m|m} + \mathbf{f}_m.$$

$$\mathbf{R}_{m+1|m} = \mathbf{A} \mathbf{R}_{m|m} \mathbf{A}^* + \mathbf{R}$$

$$\text{Analysis: } \mathbf{K}_{m+1} = \mathbf{R}_{m+1|m} \mathbf{G}^* (\mathbf{G} \mathbf{R}_{m+1|m} \mathbf{G}^* + \mathbf{R}^o)^{-1},$$

$$\bar{\mathbf{u}}_{m+1|m+1} = \bar{\mathbf{u}}_{m+1|m} + \mathbf{K}_{m+1} (\mathbf{v}_{m+1} - \mathbf{G} \bar{\mathbf{u}}_{m+1|m}),$$

$$\mathbf{R}_{m+1|m+1} = (\mathbf{I} - \mathbf{K}_{m+1} \mathbf{G}) \mathbf{R}_{m+1|m} (\mathbf{I} - \mathbf{K}_{m+1} \mathbf{G})^* + \mathbf{K}_{m+1} \mathbf{R}^o \mathbf{K}_{m+1}^*$$

Full and Partial Observation

The observation dimension M need not equal the state dimension N .

- **Partial observations** ($M \ll N$): each $\mathbf{v}_{m+1}$ senses only a few linear combinations of state components via $\mathbf{G}$; recovering unobserved components relies on dynamical coupling in $\mathbf{A}$ and repeated assimilation cycles.
- **Over-determined observations** ($M \gtrsim N$): e.g., many sensors measuring a lower-dimensional state; the same formulas apply with $\mathbf{G} \in \mathbb{C}^{M \times N}$ (Lagrangian DA).

DA Observability and Stochastic Controllability

Definition (Observability)

The pair (F, g) is *observable* if the data channel is sensitive to the state.

Definition (Stochastic controllability)

The model is *stochastically controllable* if the process noise injects uncertainty into the state $\mathbf{R} \succ 0$ (positive definite).

DA Observability and Stochastic Controllability

Example: Consider

$$\begin{aligned} u_{m+1}^{(1)} &= A_{11} u_m^{(1)} + A_{12} u_m^{(2)} + \sigma_m^{(1)}, \\ u_{m+1}^{(2)} &= A_{21} u_m^{(1)} + A_{22} u_m^{(2)} + \sigma_m^{(2)}, \end{aligned} \quad v_{m+1} = g_1 u_{m+1}^{(1)} + \sigma_{m+1}^o,$$

i.e., only the first component is directly observed (scalar $g_1 \in \mathbb{C}$). Information can improve estimates of $u^{(2)}$ *only if* $u^{(2)}$ dynamically couples to $u^{(1)}$ through $A_{12} \neq 0$ or $A_{21} \neq 0$; otherwise $u^{(2)}$ remains effectively unobservable from v_{m+1} alone.

An Application

Lighthouse tracking

A ship (u_m) at night is tracked by a lighthouse (g). No light ($g = 0$) means no tracking; no waves ($r = 0$) means perfectly predictable motion, but a slightly wrong heading can grow unless you get periodic flashes ($g \neq 0$, observability) and direction correction ($r = 0$, error correction, controllability) to update the course.

Takeaways for DA

- *Observability over time*: Partial, indirect measurements can still render the system observable if the dynamics couple hidden to measured states.
- *Stochastic controllability*: Positive-definite Q (or, more precisely, $W_c(N) \succ 0$) is essential: directions not excited by the model noise cannot be corrected unless they become observable through coupling.
- *Design hint*: When building low-order DA models, ensure (F, H) is observable and that the noise model (F, R) excites all dynamically important directions.
- Measurement noise $R^o \succ 0$ does *not* affect observability/controllability themselves, only the statistical weight in the update.

Nonlinear filtering methods

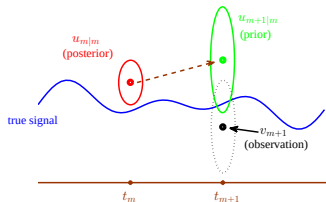
Recall the classical Kalman filter (in 1D case):

$$\text{Model: } u_{m+1} = Fu_m + \mathcal{F}_{m+1} + \sigma_{m+1},$$

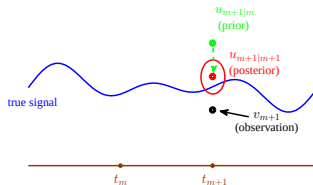
$$\text{Obs: } v_{m+1} = gu_{m+1} + \sigma_{m+1}^o.$$

- ▶ σ_{m+1} has zero mean and variance $r = \langle \sigma_{m+1} \sigma_{m+1}^* \rangle = \frac{1}{2} \sum_{j=1}^2 \langle \sigma_{j,m+1}^2 \rangle$.
- ▶ σ_m^o has zero mean and variance $r^o = \langle \sigma_m^o (\sigma_m^o)^* \rangle$.

1. Prediction (Forecast)



2. Analysis (Filtering)



The posterior mean and posterior variance are given by

$$\bar{u}_{m+1|m+1} = (1 - K_{m+1}g)\bar{u}_{m+1|m} + K_{m+1}v_{m+1},$$

$$r_{m+1|m+1} = (1 - K_{m+1}g)r_{m+1|m} \rightarrow \text{a constant as } m \rightarrow \infty.$$

where K_{m+1} is the Kalman gain

$$K_{m+1} = \frac{g r_{m+1|m}}{r^o + g^2 r_{m+1|m}} \rightarrow \text{a constant as } m \rightarrow \infty$$

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- ▶ The Kalman filter is optimal when the underlying dynamics is linear and noise is Gaussian.
- ▶ Since the dynamics is linear and Gaussian, the time evolution of the mean and variance can be written down explicitly.
- ▶ Dynamical systems in real applications are usually of dimension $O(10^6)$ – $O(10^9)$.

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Then what's the problems for the Kalman filter in real applications?

Practical issues in the Kalman filter with high-dimensional nonlinear dynamics (and/or non-Gaussian observations):

1. There is in general no analytic solutions in the forecast stage (nonlinear model).

$$\vec{u}_{m+1} = \vec{F}(\vec{u}_m) + \vec{\sigma}_{m+1}.$$

2. The posterior states have no analytic solution either (non-Gaussian statistics).

$$p(\vec{u}_{m+1}|\vec{v}_{m+1}) \sim p(\vec{u}_{m+1})p(\vec{v}_{m+1}|\vec{u}_{m+1}).$$

3. It is impossible to use numerical methods to solve either the prior or the posterior solutions (high dimensionality).

What if we still apply the Kalman filter to nonlinear problems with non-Gaussian statistics?

What if we still apply the Kalman filter to nonlinear problems with non-Gaussian statistics?

Then we need to linearize the nonlinear models.

- ▶ Dynamical properties are completely changed.
- ▶ Statistics beyond mean and covariance are ignored.

Some alternative methods are presented below.

1. The Extended Kalman Filter.

$$\vec{u}_{m+1} = \vec{F}(\vec{u}_m) + \vec{\sigma}_{m+1}$$

with nonlinear function $\vec{F}$ and $R = \langle \vec{\sigma}_m \otimes \vec{\sigma}_m^T \rangle$.

In each forecast step, a Taylor series up to order one is applied:

$$\vec{F}(\vec{u}) \approx \vec{F}(\vec{u}_{m|m}) + f_m(\vec{u} - \vec{u}_{m|m}), \quad f_m = \nabla \vec{F}(\vec{u})|_{\vec{u}=\vec{u}_{m|m}},$$

such that the prior mean and prior covariance are given by

$$\begin{aligned} \vec{u}_{m+1|m} &= \vec{F}(\vec{u}_{m|m}), \\ R_{m+1|m} &= f_m R_{m|m} f_m^T + R \end{aligned}$$

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$$\begin{aligned} \vec{u}_{m+1|m} &= \vec{F}(\vec{u}_{m|m}), \\ R_{m+1|m} &= f_m R_m f_m^T + R \end{aligned}$$

- ▶ The extended Kalman filter is a **suboptimal** filter since the evolution of both the mean and covariance contain errors introduced through the first order truncation in the Taylor series approximation
- ▶ the covariance matrix **cannot** be estimated off-line as in the linear Kalman filter problem,

$$R_{m+1|m} \not\rightarrow \text{a constant}$$

The extended Kalman filter has been used with success in some disciplines involving small dimensional state variables.

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A natural remedy is to let the prior error covariance matrix $R_{m+1|m}$ be a constant B in time and hence the expensive computational overhead is ignored.

This strategy is known as the *optimal interpolation* (OI) or the *three-dimensional variational approach* (3D-Var).

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Typically B is chosen to represent a certain asymptotic statistical quantity, which is sometimes referred to as the climatological statistics in the atmospheric and ocean science literature.

Unfortunately, the filtering skill of these static covariance approaches is suboptimal and very sensitive to the choice of B .

2. The Ensemble Kalman Filter (EnKF).

$$\vec{u}_{m+1} = \vec{F}(\vec{u}_m) + \vec{\sigma}_{m+1}$$

with nonlinear function $\vec{F}$ and $R = \langle \vec{\sigma}_m \otimes \vec{\sigma}_m^T \rangle$.

Basic idea: approximate both the prior and the posterior error covariance matrices $R_{m+1|m}$ and $R_{m+1|m+1}$ with ensemble covariance matrices around the corresponding ensemble mean $\vec{u} = K^{-1} \sum_{k=1}^K \vec{u}^k$

$$R_{m+1|m} \approx \tilde{R}_{m+1|m} = \frac{1}{K-1} U_{m+1|m} U_{m+1|m}^T$$

$$R_{m+1|m+1} \approx \tilde{R}_{m+1|m+1} = \frac{1}{K-1} U_{m+1|m+1} U_{m+1|m+1}^T$$

where

$$U_{m+1|m} = [\vec{u}_{m+1|m}^1 - \vec{u}_{m+1|m}, \vec{u}_{m+1|m}^2 - \vec{u}_{m+1|m}, \dots, \vec{u}_{m+1|m}^K - \vec{u}_{m+1|m}]^T$$

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- ▶ With these ensemble approximations, we avoid the expensive computation in the extended Kalman filter and replace it by integrating each ensemble member in time through the original model.
- ▶ In this fashion, the ensemble Kalman filter does not require any tangent approximation f_m of nonlinear operator $\vec{F}$ as in the extended Kalman filter.

The EnKF is the most widely used filtering technique in practice.

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A few subtle issues: sampling error, duplicating of the observations ...
Some modified version in practice:

- ▶ Whitaker and Hamill's ensemble square root filter (EnSRF)
- ▶ Bishop's ensemble transform Kalman filter (ETKF)
- ▶ Anderson's ensemble adjustment Kalman filter (EAKF)
- ▶ Ott et al.'s local ensemble Kalman filter (LEKF)
- ▶ ...

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2. The covariance may have band structures.
3. Though higher order statistics cannot be estimated accurately via Monte Carlo, mean and covariance are recovered reasonably well.
4. The model error is much larger than the filtering error ...

3. Particle Filter.

The main idea in particle filtering is to apply the Bayes theorem to update an ensemble of solutions (or particles) **without assuming any Gaussianity** on the prior or posterior distributions.

Thus, no explicit formulation as Kalman filter equations is used. To be more precise, consider an ensemble posterior state $\{\vec{u}_{m|m}^k \in \mathbb{R}^N\}_{k=1,\dots,K}$ of size K at time t_m with distribution function,

$$p_{m|m}(\vec{u}) = \sum_{k=1}^K w_m^k \delta(\vec{u} - \vec{u}_{m|m}^k)$$

where $\delta(x - x_0) = 1$ if $x = x_0$ and zero otherwise.

The term w_m^k determines the weight (or importance) of each ensemble member $\vec{u}_{m|m}^k$ and for $p_{m|m}$ to be a probability density function, these weights satisfy $\sum_k w_m^k = 1$ and $w_m^k \geq 0$.

Given a finite sample of the initial distribution $p_{m|m}$ through particles $\{\vec{u}_{m|m}^k\}_{k=1,\dots,K}$, one propagates these particles forward in time by solving the underlying model with initial conditions $\{\vec{u}_{m|m}^k\}_{k=1,\dots,K}$ to obtain a prior ensemble $\{\vec{u}_{m+1|m}^k\}_{k=1,\dots,K}$ and assumes that each prior particle is equally important, i.e.,

$$p_{m+1|m}(\vec{u}) = \frac{1}{K} \sum_{k=1}^K \delta(\vec{u} - \vec{u}_{m+1|m}^k)$$

Posterior states.

$$p_{m+1|m+1}(\vec{u}) = p(\vec{u}|\vec{v}_{m+1}) = \frac{p(\vec{v}_{m+1}|\vec{u})p_{m+1|m}(\vec{u})}{\int p(\vec{v}_{m+1}|\vec{u})p_{m+1|m}(\vec{u})d\vec{u}} = \sum_{k=1}^K w_{m+1}^k \delta(\vec{u} - \vec{u}_{m+1|m}^k)$$

where the weight is given by

$$w_{m+1}^k = \frac{p(\vec{v}_{m+1}|\vec{u}_{m+1|m}^k)}{\sum_{k=1}^K p(\vec{v}_{m+1}|\vec{u}_{m+1|m}^k)}.$$

Note that the Bayes formula simply re-weights the particles.

- ▶ The main issue in practice is that many particles tend to have low weights in high dimensional systems or in systems with a moderate dimensional chaotic attractor.
- ▶ If this two-step process is repeated over and over, then typically only one ensemble member will remain to have a large weight and the remaining become negligible.
- ▶ One way to avoid this ensemble collapse is to resample (known as sequential important resampling or SIR) the particles by duplicating those particles with large weights to replace those with small weights.
- ▶ There are various clever resampling strategies but none of these are applicable for high dimensional systems.

A comparison of EnKF and particle filter.

EnKF:

- ▶ efficient, deal with large dimensional systems
- ▶ only involving Gaussian statistics

Particle filter:

- ▶ able to deal with non-Gaussian statistics
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Can we improve the data assimilation algorithms by making use of the advantages in both the EnKF and the particle filter?

4. Hybrid or multiscale data assimilation.

- ▶ Strong non-Gaussianity only appears in a subspace of the entire system.
- ▶ Conditional Gaussian approximations are usually very accurate for small scale variables (where noise is large).

Consider a multi-scale filtering algorithm with forecast prior distributions

$$p^f(\mathbf{u}) = p^f(\bar{\mathbf{u}}, \mathbf{u}') \approx p^f(\bar{\mathbf{u}}) p_G^f(\mathbf{u}' | \bar{\mathbf{u}})$$

where $p_G^f(\mathbf{u}' | \bar{\mathbf{u}})$ is a Gaussian distribution conditional to $\bar{\mathbf{u}}$

$$p_G^f(\mathbf{u}' | \bar{\mathbf{u}}) = \mathcal{N}(\mathbf{u}'(\bar{\mathbf{u}}), \mathbf{R}'(\bar{\mathbf{u}})).$$

Here we assume that N_1 is sufficiently small enough that particle filters can be applied to the large scales. For a low dimensional space $\bar{\mathbf{u}}$, the marginal distribution of $\bar{\mathbf{u}}$ can be approximated by Q particles

$$p^f(\bar{\mathbf{u}}) = \sum_{j=1}^Q p_j^f \delta(\bar{\mathbf{u}} - \bar{\mathbf{u}}_j)$$

where $p_j^f \geq 0$ are particle weights such that $\sum_j p_j^f = 1$. After the forecast step, we have the following general form for the prior distribution $p^f(\mathbf{u})$

$$\begin{aligned} p^f(\mathbf{u}) = p^f(\bar{\mathbf{u}}, \mathbf{u}') &= \sum_{j=1}^Q p_j^f \delta(\bar{\mathbf{u}} - \bar{\mathbf{u}}_j) p_G^f(\mathbf{u}'^f | \bar{\mathbf{u}}_j) \\ &= \sum_{j=1}^Q p_j^f \delta(\bar{\mathbf{u}} - \bar{\mathbf{u}}_j) \mathcal{N}(\mathbf{u}'(\bar{\mathbf{u}}_j)^f, \mathbf{R}'(\bar{\mathbf{u}}_j)^f), \end{aligned}$$

which is a conditional Gaussian mixture distribution where each summand is a Gaussian distribution conditional to $\bar{\mathbf{u}}_j$.

The Gaussian mixture has already been used in data assimilation but the multi-scale method developed here is different in that conditional Gaussian distributions are applied in the reduced subspace $\mathbf{u}'$ with particle approximations only in the lower dimensional subspace $\bar{\mathbf{u}}$.

Thus the proposed multi-scale data assimilation method can be highly efficient and fast in comparison with conventional data assimilation methods which use the whole space for the filter.

The Gaussian mixture has already been used in data assimilation but the multi-scale method developed here is different in that conditional Gaussian distributions are applied in the reduced subspace $\mathbf{u}'$ with particle approximations only in the lower dimensional subspace $\bar{\mathbf{u}}$.

Thus the proposed multi-scale data assimilation method can be highly efficient and fast in comparison with conventional data assimilation methods which use the whole space for the filter.

This is now a hot cutting edge research topic with many real applications!!

Many novel (and highly non-trivial) techniques have been developed to facilitate the hybrid data assimilations. See Chen & Majda 2018 *Entropy* for a review of such ideas.

Thank you!

References and acknowledgements

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